BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE PROJECT

Submitted By

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ABSTRACT

The Bluestock Mutual Fund Analytics project focuses on building an end-to-end analytics solution for analysing mutual fund performance, investor behaviour, portfolio allocation, and risk metrics using Python, SQLite, and Power BI. The project includes ETL pipeline development, data cleaning, exploratory data analysis, performance metric computation, advanced risk analytics, and interactive dashboard creation.

The project aims to provide meaningful investment insights through financial analytics techniques such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha-Beta analysis, Value at Risk (VaR), Conditional Value at Risk (CVaR), and portfolio concentration analysis. The final solution enables interactive analysis of fund performance, SIP growth, investor demographics, and sector allocation trends.

1. INTRODUCTION

Mutual funds are one of the most popular investment instruments for retail and institutional investors. With increasing participation in the Indian financial market, analysing fund performance and investor behaviour has become essential for investment decision-making.

This project was developed to create a complete analytics platform capable of:

- Analysing mutual fund NAV trends
- Tracking SIP inflows and folio growth
- Evaluating fund risk-adjusted returns
- Understanding investor demographics
- Building interactive dashboards for business insights

The project integrates data engineering, data analytics, financial analytics, and dashboard visualisation into a single end-to-end workflow.

2. OBJECTIVES

The major objectives of the project are:

- Build an ETL pipeline for mutual fund datasets
- Clean and validate financial datasets
- Design a SQLite database schema
- Perform exploratory data analysis (EDA)
- Compute mutual fund performance metrics
- Develop advanced financial risk analytics
- Create an interactive Power BI dashboard
- Generate actionable investment insights

3. TECHNOLOGIES USED

Technology	Purpose
Python	Data processing and analytics
Pandas	Data manipulation

Technology	Purpose
NumPy	Numerical computations
Matplotlib	Data visualisation
Seaborn	Statistical charts
SQLite	Database management
SQLAlchemy	Database connectivity
Power BI	Interactive dashboard
Git & GitHub	Version control

4. DATASETS USED

Dataset	Description
Fund Master	Scheme details and metadata
NAV History	Daily NAV history of funds
AUM by Fund House	Assets under management
Monthly SIP Inflows	SIP trend analysis
Category Inflows	Category-wise fund flows
Industry Folio Count	Investor folio growth
Scheme Performance	Fund performance metrics
Investor Transactions	Investor transaction history
Portfolio Holdings	Sector and holdings allocation
Benchmark Indices	Nifty benchmark performance

5. ETL PIPELINE

The ETL pipeline was developed using Python to automate:

- Data ingestion
- Data cleaning
- Data validation

- Data transformation
- Database loading

The pipeline reads raw CSV files, applies preprocessing operations, and stores cleaned datasets in SQLite.

Key ETL tasks:

- Missing value handling
- Duplicate removal
- Datetime conversion
- Data standardisation
- NAV forward filling
- Schema validation

6. DATABASE DESIGN

A SQLite star schema was implemented for structured analytics.

The schema includes:

- Dimension tables
- Fact tables
- Primary and foreign keys

Main tables:

- dim_fund
- dim_date
- fact_nav
- fact_transactions
- fact_performance
- fact_aum

The database supports analytical SQL queries and dashboard integration.

7. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to identify trends and patterns in mutual fund investments.

The analysis included:

- NAV trend analysis
- SIP inflow analysis
- Category inflow heatmaps
- Investor demographics
- Geographic investment distribution
- Sector allocation analysis
- Folio growth trends

Key visualisations were generated using Plotly, Matplotlib, and Seaborn.

8. PERFORMANCE ANALYTICS

Several financial performance metrics were calculated for all mutual fund schemes.

CAGR (Compound Annual Growth Rate)

Used to measure long-term annualised growth of mutual funds.

Sharpe Ratio

Used to evaluate risk-adjusted return performance.

Sortino Ratio

Focused on downside risk-adjusted performance.

Alpha and Beta

Used to compare fund performance against benchmark indices.

Maximum Drawdown

Measured the maximum historical decline from peak NAV.

A composite fund scorecard was developed using:

- Return rank
- Sharpe Ratio
- Alpha
- Expense ratio
- Drawdown

9. ADVANCED ANALYTICS

Advanced financial analytics were implemented for deeper investment analysis.

Historical VaR & CVaR

Risk metrics were calculated using historical daily return distributions.

Rolling Sharpe Ratio

A rolling 90-day Sharpe Ratio was computed to analyse performance stability.

Investor Cohort Analysis

Investors were grouped based on their first investment year.

SIP Continuity Analysis

Investors with irregular SIP behaviour were identified.

Fund Recommendation System

A rule-based recommender system was created based on:

- Risk appetite
- Sharpe Ratio
- Risk grade

Sector HHI Concentration

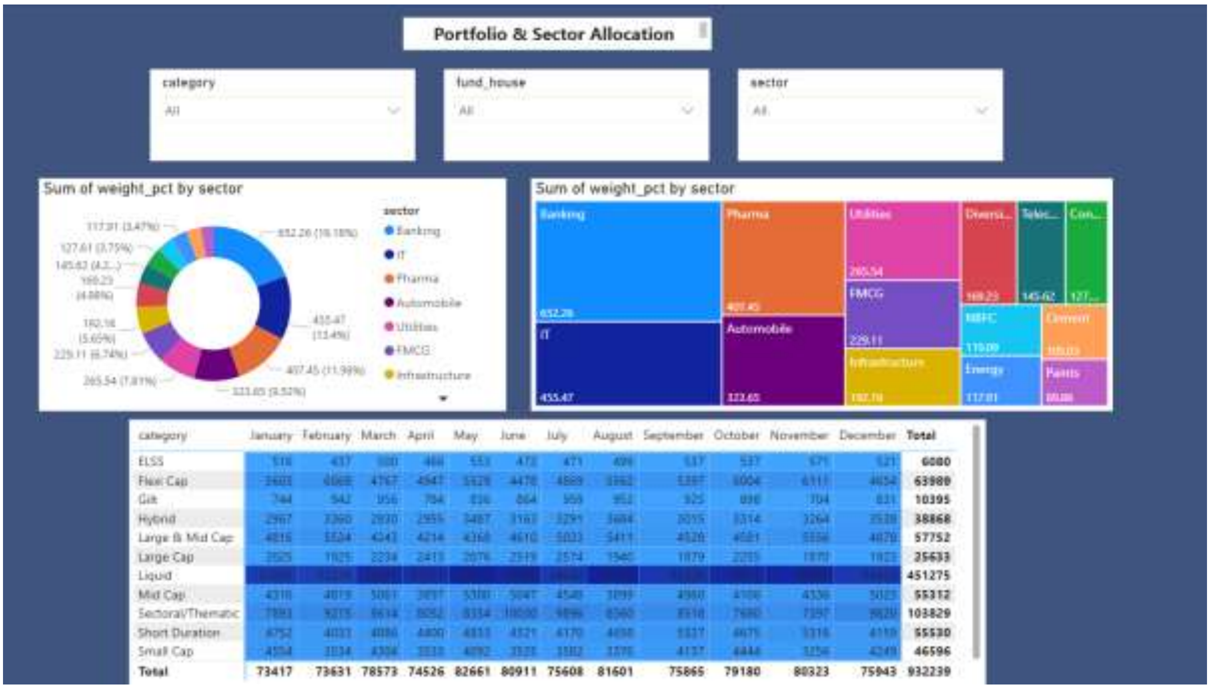
Portfolio concentration risk was measured using the Herfindahl-Hirschman Index (HHI).

10. DASHBOARD DEVELOPMENT

An interactive Power BI dashboard was developed with multiple analytical pages.


Mutual Fund Performance
39M
Sum of aum_crore
940K
Sum of sip_inflow_crore
0.15
Avg Cagr
26.12
Total Folios
0.52
Avg sharp_ratio
fund_house
All
category
All
NAV Growth Trend
scheme_name
date
Sum of nav
Monthly SIP Inflow Trend
month
Industry Folio Growth
Year
55
63
101
197


Fund Performance & Risk
fund_house
All
risk_grade
High
Low
Moderate
Moderately High
Very High
scheme_name
All
Sum of fund_score by scheme_name
HDFC Short Term Debt Fund - Regular - ...
Nippon India Gilt Securities Fund - Regu...
SBI Magnum Gilt Fund - Regular Plan - ...
ABSL Liquid Fund - Regular - Growth
UTI Mid Cap Fund - Regular - Growth
Kotak Emerging Equity Fund - Regular - ...
Kotak Liquid Fund - Regular - Growth
Sum of fund_score
Sum of nav by benchmark
CRISIL Liquid Fund AI Index
NIFTY 100 TRI
NIFTY Midcap 150 TRI
NIFTY 500 TRI
BSE 250 SmallCap TRI
NIFTY 50 TRI
NIFTY Large Midcap 250 ...
NIFTY Midcap 50 TRI
CRISIL Dynamic Gil...
CRISIL Short Term Bond I...
Sum of sharpe_ratio, Sum of cagr and Sum of alpha
Sum of cagr



Dashboard sections include:

- Executive Overview
- Fund Performance Analytics
- Investor Demographics
- Portfolio Analytics
-

Features:

- KPI cards
- Interactive slicers
- Dynamic charts
- Benchmark comparisons
- Risk analytics

11. KEY INSIGHTS

- SIP inflows showed continuous growth between 2022 and 2025.
- Equity funds dominated category-wise inflows.
- Funds with higher Sharpe Ratios demonstrated stronger risk-adjusted returns.
- Investor participation increased significantly during 2024–2025.
- Some portfolios showed high sector concentration risk based on HHI values.
- Rolling Sharpe analysis revealed varying fund stability over time.
- Folio counts increased substantially, indicating rising retail participation.

12. CHALLENGES FACED

- Handling missing NAV values during holidays and weekends
- Managing large financial datasets
- Building proper Power BI relationships
- Handling date hierarchy issues
- Implementing advanced financial metrics correctly

CONCLUSION

The Bluestock Mutual Fund Analytics project successfully implemented a complete financial analytics workflow covering ETL processing, database management, exploratory analysis, risk analytics, and dashboard visualisation.

The project demonstrates practical applications of:

- Data Analytics
- Financial Analytics
- SQL
- Dashboarding
- Business Intelligence

The final solution provides meaningful investment insights and supports data-driven mutual fund analysis.

FUTURE SCOPE

Possible future enhancements include:

- Real-time NAV API integration
- Machine learning-based fund recommendation

- Monte Carlo simulation
- Portfolio optimisation models
- Automated reporting systems
- Streamlit web application deployment

REFERENCES

- AMFI India
- NSE India
- Pandas Documentation
- Power BI Documentation
- SQLite Documentation
- Python Financial Analytics Resources